

ALL ABOUT OUR NEED

INTRODUCTION

Podcasts have become a widely used medium for sharing knowledge and entertainment. Many listeners seek focused, topic-driven discussions, yet casual, off-topic conversations often detract from this experience. The challenge lies in classifying conversations into casual and non-casual segments effectively. This research project aims to develop a classification method that can distinguish between casual and non-casual conversations. By utilizing Large Language Models (LLMs) and zero-shot learning, the approach avoids the need for large labeled datasets, enhancing scalability and adaptability across diverse podcast formats

Problem Definition This project seeks to create a robust classification method to accurately identify and differentiate casual and non-casual segments in podcast dialogues leveraging Large Language Models.

CORE IDEA FROM THIS RESEARCH PAPER:

Link: <https://arxiv.org/abs/2305.08377>

Text Classification with Large Language Models (LLMs): Text classification assigns labels to texts, but LLMs like GPT-3 face challenges compared to fine-tuned models.

The Core Idea - CARP Framework (Clue and Reasoning Prompting)

Main Approach:

- **Finding Clues:** Identify important keywords, tones, or references within the text.
- **Reasoning Process:** Use those clues to make more informed decisions on text classification.
- **Key Benefit:** Combines LLMs' general knowledge with fine-tuned models to find relevant examples and enhance the classification process.

Detailed Process:

- **Clue Identification:** The model highlights essential clues in the text (e.g., keywords, tone).
- **Reasoning:** It uses these clues to better understand the text context, considering nuanced aspects like irony or sentiment strength.
- **Final Decision:** The model makes the classification decision based on the clues and reasoning process.

Example: For a sarcastic review, CARP identifies phrases like "snappy screenplay" to conclude a positive sentiment.

EXAMPLE

This is an overall sentiment classifier for movie reviews.
First, list CLUES (i.e., keywords, phrases, contextual information, semantic relations, semantic meaning, tones, references) that support the sentiment determination of input..
Second, deduce the diagnostic REASONING process from premises (i.e., clues, input) that supports the INPUT sentiment determination (Limit the number of words to 130).
Third, based on clues, reasoning and input, determine the overall SENTIMENT of INPUT as Positive or Negative.

INPUT: press the delete key

CLUES: delete key

REASONING: The phrase "delete key" implies an action of removing something, which could be interpreted as a negative sentiment.

SENTIMENT: Negative ✓

(c) CARP Zero-shot

Github link of code implementation for sentiment classification
<https://github.com/ShannonAI/GPT-CLS-CARP>

OUR DESIRED MODIFIED CODE :

- **Input**
 - Raw podcast transcript containing both casual, off-topic discussions and relevant content.
- **Output**
 - Classified transcript where each segment is labeled as either Relevant or Casual.

Incorporating paper idea example

Suppose it is a podcast involving Elon musk. We first ask LLM about his family, food, and hobbies as the first prompt. So now we are creating an environment for LLM. Then on the subsequent prompt we provide the conversations to classify. Now we ask them to classify.

FINAL MAIN REQUIREMENTS:

- Make the existing implementation working for sentiment analysis only for zero shot (**Challenge: name of dataset used is given in the paper which is common, but not the actual format of the dataset they used, when we tried a few it gave attribute error**)
- Modify it for casual conversation identification in podcast
- Incorporate evaluation metrics and get the results (here the evaluation metrics is not simply precision, recall)